

CHE 891 s302

Machine Learning for Chemical Process Engineering

Spring 2026

Course Mode

In-person lecture/recitation

Course Dates

Tuesday Feb 17, 2026 — Friday March 20, 2026

Lecture/recitation

Tuesdays 12:40 pm – 2:00 pm, 2245 EB; **Fridays** 11:00 am – 12:20 pm, 2245 EB

Instructor

Scott Calabrese Barton 3243 EB

+1-517-355-0222, scb@msu.edu, <https://msu.zoom.us/my/scbmsu>

Office Hours

Tuesdays 2 – 3 pm, 2245 EB or by appointment

Course Description

This course applies foundational supervised and unsupervised machine learning approaches to chemical processes. Process knowledge and industrial datasets are integrated with statistical learning for real-world deployment challenges. Topics include unsupervised learning and clustering for identifying operational regimes, dimensionality reduction methods like principal component analysis and partial least squares for process monitoring and soft sensor development, and neural networks including recurrent and autoencoder architectures for dynamic modeling. A central theme is the spectrum of hybrid modeling approaches — from online bias correction to physics-informed neural networks — that embed process knowledge into data-driven models. Final projects involve developing hybrid physics-machine learning models for chemical processes.

Prerequisites

- Introductory machine learning course in Python, covering data preprocessing, supervised learning (regression/classification), model evaluation, and hyperparameter tuning. We will utilize mainly `scikit-learn` and `TensorFlow` in this course.
- Chemical engineering fundamentals including process dynamics and control

Optional Texts

- A. Géron, *Hands-On Machine Learning with Scikit-Learn, Keras & Tensorflow*, 3rd Edition, O'Reilly, 2023, ISBN 978-1-098-12597-4
 - github: <https://github.com/ageron/handson-ml3>

- W. McKinney, *Python for Data Analysis*, 3rd Edition, O'Reilly, 2022. ISBN 9781098104030
 - [web site](#), [github](#)

Electronic Resources

- Course Materials Web Site: Desire2Learn <https://d2l.msu.edu/d2l/home/2550756>
- Computation: [Python Version 3.12](#), with `numpy`, `scipy`, `matplotlib`, `pandas`, `scikit-learn` and `tensorflow` packages.
- Discussion: Slack or Teams, TBD.

Grading

Grade Item	Weight
Homework	20%
Project	40%
Class participation	40%

Course grades will be determined as follows:

85-100	80-85	70-80	60-70
4.0	3.5	3.0	2.5

Appropriate Use of Generative AI

In this course, generative AI tools (i.e. ChatGPT, Claude, Copilot, etc.) are mainly helpful as a tool to troubleshoot code, a reference for language syntax, and to check for grammar and style. All other uses are strongly discouraged, especially for implementing the ML libraries (TensorFlow, sklearn) that are the subject of the course. You are required to cite in your work if AI is used in these ways. Instead, please construct your own code and write reviews and summaries on your own.

Schedule

Week 1

Topics

- Unsupervised learning and clustering
- Dimensionality Reduction and feature selection
- Partial Least Squares for relating process variables to quality metrics

Readings

- Géron Chapter 8
- S. J. Qin, S. Guo, Z. Li, L. H. Chiang, I. Castillo, B. Braun and Z. Wang, "Integration of process knowledge and statistical learning for the Dow data challenge problem", *Computers &*

Chemical Engineering, **153**, 107451 (2021). [doi:10.1016/j.compchemeng.2021.107451](https://doi.org/10.1016/j.compchemeng.2021.107451)

- §1-3

Supporting

- S. Wold, M. Sjöström and L. Eriksson, "PLS-regression: a basic tool of chemometrics", *Chemometrics and Intelligent Laboratory Systems*, **58**, 109-130 (2001). doi:[10.1016/S0169-7439\(01\)00155-1](https://doi.org/10.1016/S0169-7439(01)00155-1)

Assignment

Dow data cleaning and filtering with process knowledge

Week 2

Topics

- Variable selection methods: Least Angle Regression, stepwise methods
- t-SNE and UMAP for visualization of high-dimensional process spaces
- Bias detection and correction for model drift (Online Bias Learning)

Readings

- Géron Chapter 4: Regularized Linear Models
- S. J. Qin, S. Guo, Z. Li, L. H. Chiang, I. Castillo, B. Braun and Z. Wang, "Integration of process knowledge and statistical learning for the Dow data challenge problem", *Computers & Chemical Engineering*, **153**, 107451 (2021). [doi:10.1016/j.compchemeng.2021.107451](https://doi.org/10.1016/j.compchemeng.2021.107451)
 - §4-7

Supporting

- B. Efron, T. Hastie, I. Johnstone and R. Tibshirani, "Least angle regression", *Ann. Statist.*, **32**, (2004). doi:[10.1214/009053604000000067](https://doi.org/10.1214/009053604000000067)

Assignment

Modeling Dow data using two methods

Week 3 (after Spring Break) Neural Networks for Process Systems

Topics

- Feedforward NNs for dynamic process prediction
- Neural network training with explicit gradient computation
- Custom loss functions with physics-based soft constraints

Assignment

Project: Build a physics-informed neural network for a Van de Vusse CSTR or similar

Readings

- D. C. Psychogios and L. H. Ungar, "A hybrid neural network-first principles approach to process modeling", *AIChE Journal*, **38**, 1499–1511 (1992). doi:[10.1002/aic.690381003](https://doi.org/10.1002/aic.690381003)

Supporting

- J. G. Van De Vusse, "Plug-flow type reactor versus tank reactor", *Chemical Engineering Science*, **19**, 994–996 (1964). doi:[10.1016/0009-2509\(64\)85109-5](https://doi.org/10.1016/0009-2509(64)85109-5)
-

Week 4: Physics-Informed Neural Networks

Topics

- The PINN loss function
- Automatic differentiation: computing physics residuals from NN outputs
- Forward problems (predicting states) vs. inverse problems (estimating unknown parameters)
- State estimation: inferring unmeasured process states when constitutive equations are incomplete

Lab/Assignment

Project continued

Readings

- Raissi, M., Perdikaris, P., & Karniadakis, G. E. (2019). "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations." *Journal of Computational Physics*, 378, 686–707. DOI:[10.1016/j.jcp.2018.10.045](https://doi.org/10.1016/j.jcp.2018.10.045)
- Velioglu, M., et al. (2025). "Physics-informed neural networks for dynamic process operations with limited physical knowledge and data." *Computers & Chemical Engineering*, 192, 108899. DOI:[10.1016/j.compchemeng.2024.108899](https://doi.org/10.1016/j.compchemeng.2024.108899)

Supporting:

- Karniadakis, G. E., et al. (2021). "Physics-informed machine learning." *Nature Reviews Physics*, 3, 422–440. DOI:[10.1038/s42254-021-00314-5](https://doi.org/10.1038/s42254-021-00314-5)
- Wu, Z., et al. (2020). "Machine learning-based predictive control of nonlinear processes. Part I: Theory." *AIChE Journal*, 66(11), e16729. DOI:[10.1002/aic.16729](https://doi.org/10.1002/aic.16729)